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Notifications 

3. Solving inhomogeneous systems

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Recitation due Feb 14, 2024 20:30 IST

Determinants and inverses (External resource)

(1.0 points possible)

Determinants and inverses

The formula for finding the determinant of a square matrix becomes very complicated fc finding an expression for its inverse is even harder. Thankfully there are very simple cor these operations for us.

- We can calculate the determinant $\det(\mathbf{A})$ of any square matrix $\mathbf{A}$ using the foll

```
d = det(A)
```

- Recall that if $\det(\mathbf{A}) \neq 0$ then we can calculate the inverse $\mathbf{B} = \mathbf{A}^{-1}$. The MATL

```
B = inv(A)
```

- Given a linear system $\mathbf{Ax} = \mathbf{b}$, where $\det(\mathbf{A}) \neq 0$, we can find the solution $\mathbf{x} =$ commands:

```
x = inv(A)*b
%This is doing two things. Firstly it calculates the inverse of A, and
%
x = A\b
%This is the recommended way to solve a linear system. It is faster th
%
```

In this problem we will consider solving a large system of equations written in matrix for random 10×10 matrix $\mathbf{A}$ and a random 10×1 column vector $\mathbf{b}$. Use MATLAB to chec the linear system.

Script ?

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MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 % Firstly we will generate a random 10x10 matrix A and a random RHS
2 % You should use the command X=rand(M,N), where M is the number of
3 %
4 A = rand(10,10);
5 b = rand(10,1)
6 %
7 % Now calculate the determinant of A and assign its value to a vari
8 %
9 d = det(A)
10 %
11 % Now solve the linear system using the function inv() or \ to find
12 %
13 x=inv(A)*b
14 %
15 % You can check for yourself that this does indeed solve the system
```

▶ Run Script

?

Assessment: All Tests Passed

Submit ?

✔ Are A and b defined correctly?

✔ Is the solution vector correct?

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✔ d is correct

3. Solving inhomogeneous systems

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